

A Fast Pessimistic Diagnosis Algorithm for Generalized Hypercube Multicomputer Systems

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Abstract

This work presents a system-level diagnosis algorithm for a generalized hypercube which is an attractive variance of a hypercube. The proposed algorithm is based on the PMC model and can isolate all faulty nodes to within a set that contains at most one fault-free node. If the total number of nodes to be diagnosed in a generalized hypercube is N , the proposed algorithm can run in $O(N \log N)$ time, and, being superior to Yang's algorithm proposed in 2004, it can diagnose not only a hypercube but also a generalized hypercube.

1 Introduction

Fault-tolerance computing is very important for a massively parallel processing system and the reliability of processors in it is therefore becoming an important issue for designing the system. In order to achieve high system reliability and availability, a faulty processor (node) when it is found should be replaced by a fault-free processor. Within a multiprocessor system, the technique of identifying faulty nodes by constructing tests on the nodes and interpreting the test outcomes is known as *system-level diagnosis* [6].

Several diagnosis models were proposed to identify the faulty nodes. One important approach first proposed by Preparata, Metze and Chien is called the *PMC diagnosis model* [6]. Every node (tester) in a system carries out tests on its neighbors via the edges (communication links) between them; when a node tests another node, the tester declares the tested node to be 'fault-free' or 'faulty' depending on the test response, and this evaluation is reliable if and only if the tester is 'fault-free'.

There are two fundamental different strategies for system-level diagnosis. One is the traditional *precise diagnosis* strategy by which all nodes can be identified correctly, namely, all fault-free nodes

diagnosed as 'fault-free' and all faulty nodes diagnosed as 'faulty' [6]. The other diagnosis strategy is *pessimistic diagnosis* strategy which underestimates the diagnostic capability of a system whenever the number of faulty processors exceeds the precise diagnosability of the system [4], [12].

A new diagnosis strategy proposed by Somani and Peleg in 1996 is known as the t/k diagnosis [8]. At the expense of lower diagnosis demand the basic idea behind the t/k diagnosis is to enhance the self-diagnosing capability of multicomputer systems [3], [7], [8]. Specifically, a system is t/k -diagnosable if the number of faulty nodes is bounded by t and all faulty nodes can be isolated to within a set in which at most k nodes are false-alarm. Notably, the precise diagnosis is the special case of the t/k diagnosis with k is 0. If k is 1, the t/k diagnosis is equal to another diagnosis strategy, the pessimistic diagnosis. Comparing the precise diagnosis with the pessimistic diagnosis, the latter one can enhance the diagnosis capability of multicomputers at the cost of slightly lower diagnosis demand.

Yang proposed an $O(N \log N)$ time diagnosis algorithm (CD-DIAG) for a hypercube multicomputer system of N processors (nodes) [11]. A generalized hypercube (GH for short) is a generalization of a hypercube, and is also a kind of cube which can be developed irregularly. Under the same dimension, the degree of each node of a generalized hypercube is not less than the one of a hypercube. This work derives an $O(N \log N)$ time diagnosis algorithm for a N -processor generalized hypercube multicomputer system. Naturally, the provided algorithm is remarkably superior to the CD-DIAG algorithm proposed by Yang in 2004.

The topological structure of a multicomputer system (an interconnection network) can be modeled by a graph [10] of which the vertices and edges of a graph correspond to nodes and links of a multicomputer system (an interconnection network), respectively. Therefore, this work regards a generalized hypercube as a graph and the analysis adopts the terms used by some text books of graph theory [2], [9], [10].

The rest of the paper is organized as follows.

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Section 2 describes background and notations to be used in this paper. Section 3 addresses the cycle decomposition properties of a generalized hypercube. Section 4 presents a pessimistic diagnosis algorithm for a generalized hypercube, followed by the correctness proof and the time complexity analysis. Conclusions are finally made in Section 5.

2 Background and Notations

This section discusses the structure of a generalized hypercube, PMC model, and some properties and notations about them.

2.1 Generalized Hypercube

A generalized hypercube graph is the underlying graph of a generalized hypercube network which is proposed by Bhuyan and Agrawal in 1984 for building a massively parallel computer system [1]. In the following, each node (each edge) of a GH represents a processor (a bidirectional communication link between two processors).

Let $GH(m_1, m_2, \dots, m_r)$ denote a r -dimensional generalized hypercube graph (network) of size $m_1 \times m_2 \times \dots \times m_r$, where $m_i \geq 2$ for $1 \leq i \leq r$. Each node of a $GH(m_1, m_2, \dots, m_r)$, totally $N = m_1 \times m_2 \times \dots \times m_r$ of them, is assigned a r -digit identifier $x_1x_2\dots x_r$, where $x_i \in [0, m_i-1]$ for $1 \leq i \leq r$. Two nodes in a $GH(m_1, m_2, \dots, m_r)$ are adjacent if and only if their identifiers differ at exactly one digit position. An i -edge connects two adjacent nodes whose identifiers differ at i position, where $1 \leq i \leq r$. Two nodes are i -neighbors if they connected by a i -edge. Figure 1 shows the structure of a $GH(4, 3, 2)$. Each node in a $GH(m_1, m_2, \dots, m_r)$ has degree $\sum_{i=1}^r (m_i-1)$. Notably, the order of dimensions can be exchanged and the structure is not changed. For instance, a $GH(4, 3, 2)$ is isomorphic to a $GH(2, 3, 4)$ [9].

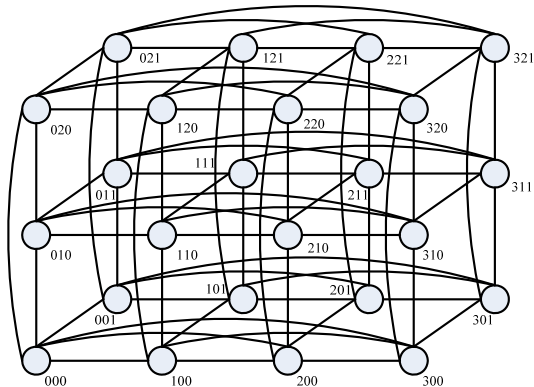


Figure 1. The structure of a $GH(4, 3, 2)$.

Let $G = (V(G), E(G))$ be the underlying graph where $V(G)$ ($E(G)$) represents the set of nodes (edges) in G . Also let $(u, v) \in E(G)$ denote the edge between two nodes u, v in G . A cycle that contains every node in G exactly once is a *Hamiltonian cycle* of G , denoted as $HC(G)$. A graph is *Hamiltonian* if it has a Hamiltonian cycle. Normally, for a graph G and a node $u \in V(G)$, let $N(u) = \{v \in V(G) : (u, v) \in E(G)\}$ ($N(S) = \{v \in V(G) : (u, v) \in E(G), u \in S \text{ and } S \subset V(G)\}$) stand for the set of neighbors of node u (node set S) in G . For other fundamental graph-theoretic terminologies, the reader can refer to [2], [9], [10].

2.2 PMC Model

Considering a GH as a multicomputer system, if one node is capable, it can be used to test a neighbor node (there is a direct communication link between them). Thus, each underlying edge is regarded as a bidirectional communication link between adjacent nodes in the GH for running a diagnosis algorithm.

Let u and v be a pair of adjacent nodes of G . The outcome of the test conducted by u on v , denoted as $\sigma(u, v)$, is 0 (or 1) if u evaluates v as ‘fault-free’ (or as ‘faulty’). The collection of all test outcomes, denoted as σ , is referred to as a *syndrome*. This work adopts the PMC model, which assumes that a fault-free node can correctly evaluate all its neighbors and the outcome of a test conducted by a faulty node is completely unreliable [6]. Table 1 lists the assumptions made in the PMC model.

Table 1. The Assumption in the PMC Model

Tester node u	Tested node v	Test outcome $\sigma(u, v)$
Fault-free	Fault-free	0
Fault-free	Faulty	1
Faulty	Fault-free	0 or 1
Faulty	Faulty	0 or 1

2.3 Properties and Notations

According to the definition of the PMC model, some properties and notations are listed below.

Lemma 1 [11]. Let G be a graph, σ be a syndrome on G , and u, v be a pair of adjacent nodes in G .

- (i) If $\sigma(u, v) = \sigma(v, u) = 0$, then u and v are of the same status.
- (ii) If $\sigma(u, v) = 1$ or $\sigma(v, u) = 1$, then u and v are of different statuses.

The PMC model gives a syndrome σ on a graph G . A cycle is a σ -zero cycle within G if the test

outcomes are all 0. Every node on a σ -zero cycle is a σ -zero node in the cycle. A cycle is σ -nonzero cycle within G if the outcome of at least one test is 1. Every node in a σ -nonzero cycle is a σ -nonzero node in the cycle. Based on the PMC model, the following lemma can be obtained.

Lemma 2 [11]. Assume C is a cycle in a graph G , and σ is a syndrome on G .

- (i) If C is σ -zero, then all nodes on C are of the same status, namely either all fault-free or all faulty.
- (ii) If C is σ -nonzero, then at least one node in C is faulty.

3 Some Properties of a GH

This section addresses some properties of a generalized hypercube, which are useful for designing the diagnosis algorithm for a GH.

Lemma 3. If $S = \{u, v\} \subset V(\text{GH}(m_1, m_2, \dots, m_r))$ and $m_j = \max\{m_1, m_2, \dots, m_r\}$, then $|N(S)| \geq 2\sum_{i=1, \dots, r} (m_i - 1) - m_j$.

Proof. By the well-known inclusive-exclusive principle, $|N(S)|$ can be computed as follows.

$$\begin{aligned} |N(S)| &= |(N(u) - S) \cup (N(v) - S)| \\ &= |N(u) - S| + |N(v) - S| \\ &\quad - |(N(u) - S) \cap (N(v) - S)| \end{aligned}$$

Let $d(u, v)$ represent the distance between u and v . There are three cases to be considered.

Case 1. $d(u, v) = 1$. $|N(u) - S| = |N(v) - S| = \sum_{i=1, \dots, r} (m_i - 1) - 1$ and $|(N(u) - S) \cap (N(v) - S)| \leq m_j - 2$. Thus, $|N(S)| \geq 2\sum_{i=1, \dots, r} (m_i - 1) - m_j$.

Case 2. $d(u, v) = 2$. $|N(u) - S| = |N(v) - S| = \sum_{i=1, \dots, r} (m_i - 1)$ and $|(N(u) - S) \cap (N(v) - S)| = 2$. Thus, $|N(S)| = 2\sum_{i=1, \dots, r} (m_i - 1) - 2$.

Case 3. $d(u, v) > 2$. $|N(u) - S| = |N(v) - S| = \sum_{i=1, \dots, r} (m_i - 1)$ and $|(N(u) - S) \cap (N(v) - S)| = 0$. Thus, $|N(S)| = 2\sum_{i=1, \dots, r} (m_i - 1)$.

Consequently, $|N(S)| \geq 2\sum_{i=1, \dots, r} (m_i - 1) - m_j$. $\square$

Lemma 4. A $\text{GH}(m_1, m_2, \dots, m_r)$ is Hamiltonian.

Proof. This lemma is proved by induction on dimension r . A $\text{GH}(m_1)$ is Hamiltonian since it is a complete graph of m_1 nodes and every edge in the $\text{HC}(\text{GH}(m_1))$ is a 1-edge. Assume that a $\text{GH}(m_1, m_2, \dots, m_k)$ is Hamiltonian for $2 \leq k < r$ and every edge in a $\text{HC}(\text{GH}(m_1, m_2, \dots, m_k))$, which connects two distinct nodes in two different $\text{GH}(m_1, m_2, \dots, m_{k-1})$ s, is a k -edge.

Regarding every $\text{GH}(m_1, m_2, \dots, m_k)$ as a supernode, a $\text{GH}(m_1, m_2, \dots, m_{k+1})$ is a complete graph of m_{k+1} supernodes and has a Hamiltonian

cycle of m_{k+1} supernodes, denoted by $\text{HC}^*(m_{k+1})$. A Hamiltonian cycle in a $\text{GH}(m_1, m_2, \dots, m_r)$ can be constructed by merging each $\text{HC}(\text{GH}(m_1, m_2, \dots, m_k))$ according to the connection of the $\text{HC}^*(m_{k+1})$. Two cases have to be considered.

Case 1. $m_{k+1} = 2$. Obviously, there exists a $\text{HC}(\text{GH}(m_1, m_2, \dots, m_k))$ in each of two $\text{GH}(m_1, m_2, \dots, m_k)$ s. Choose two k -neighbors u and v in one $\text{HC}(\text{GH}(m_1, m_2, \dots, m_k))$ and u' (v') which is a $(k+1)$ -neighbor of u (v) in the other $\text{HC}(\text{GH}(m_1, m_2, \dots, m_k))$. A $\text{HC}(\text{GH}(m_1, m_2, \dots, m_{k+1}))$ can be built by removing two k -edges (u, v) and (u', v') and adding two $(k+1)$ -edges (u, u') and (v, v') .

Case 2. $m_{k+1} > 2$. There exists a $\text{HC}(\text{GH}(m_1, m_2, \dots, m_k))$ in each $\text{GH}(m_1, m_2, \dots, m_k)$ in the $\text{GH}(m_1, m_2, \dots, m_{k+1})$. Thus, there are totally m_{k+1} $\text{HC}(\text{GH}(m_1, m_2, \dots, m_k))$ s. Similarly, merge every two adjacent $\text{HC}(\text{GH}(m_1, m_2, \dots, m_k))$ s as described in Case 1. Therefore, a $\text{HC}(\text{GH}(m_1, m_2, \dots, m_{k+1}))$ can be established by removing $2m_{k+1}$ k -edges and adding $2m_{k+1}$ $(k+1)$ -edges. $\square$

The main idea of this work is briefly described as follows. Based on Lemmas 1 and 2, decompose a GH into subcubes with the same structure, and according to Lemma 4, construct a Hamiltonian cycle in each subcube before diagnosing started. To achieve the goal, first, an algorithm to decompose a GH into proper subcubes is applied. Second, according to Lemma 4, a Hamiltonian cycle in each decomposed subcube is built. Each Hamiltonian cycle then decides if it is a σ -zero cycle or a σ -nonzero cycle by self-diagnosing. Third, a diagnosis algorithm is provided to adopt every σ -zero cycle conducted by all fault-free nodes to diagnose any σ -nonzero cycle which is adjacent to the σ -zero cycle. Eventually, each node in any undiagnosed σ -nonzero cycle, which is not adjacent to any σ -zero cycle, can be diagnosed in finite steps.

Let GH^* represent a decomposed proper subcube of a GH. Also let f , O_{GH^*} and N_{GH^*} denote the number of faulty nodes, the order of a GH^* and the total number of GH^* s in the GH, respectively. The most important issue is to determine the values of O_{GH^*} and N_{GH^*} . Naturally, $O_{\text{GH}^*} = |V(\text{GH}^*)|$ and $N_{\text{GH}^*} = |V(\text{GH})|/|V(\text{GH}^*)|$, and O_{GH^*} and N_{GH^*} should be greater than f to guarantee that there exists at least one fault-free σ -zero cycle. In other words, O_{GH^*} and N_{GH^*} decide the maximum value of f .

Since m_i s for $1 \leq i \leq r$ in a $\text{GH}(m_1, m_2, \dots, m_r)$ may not be the same, the CD-DIAG proposed by Yang can not be immediately applied to the $\text{GH}(m_1, m_2, \dots, m_r)$. Moreover, the cycle

decomposition method provided by Yang also cannot be used in a GH. In the rest of this paper, without loss of generality, assume $m_1 \geq m_2 \geq \dots \geq m_r$ in a $GH(m_1, m_2, \dots, m_r)$ where $m_i \geq 2$ for $1 \leq i \leq r$. To decompose a GH into GH^* s such that O_{GH^*} is closed to N_{GH^*} is the partition decision problem which is known as a NP-complete problem [5]. A heuristic algorithm is derived for decomposing a GH into GH^* s for maximizing f . According to Lemma 3, f should be less than $2\sum_{i=1..r}(m_i-1) - m_j$. If not, at least two nodes may be isolated and they cannot be correctly diagnosed. Consequently, $f < \min\{O_{GH^*}, N_{GH^*}, 2\sum_{i=1..r}(m_i-1) - m_j\}$.

Restated, after applying the following decomposition algorithm, every σ -zero cycle is constructed by fault-free nodes. Let $GH^* \times m_j$ indicate a GH with $|V(GH^*)| \times m_j$ nodes constructed by m_j GH^* s in a $GH(m_1, m_2, \dots, m_r)$, where $1 \leq i \leq r$. Algorithm DECOMP_GH is formally stated in the following.

Algorithm DECOMP_GH

Input: A $GH(m_1, m_2, \dots, m_r)$ where $m_1 \geq m_2 \geq \dots \geq m_r$

Output: A GH^* , f , O_{GH^*} and N_{GH^*}

Initialization: $N_{GH^*} \leftarrow 1$, $O_{GH^*} \leftarrow \prod_{i=1..r} m_i$, $h \leftarrow 1$, and $t \leftarrow r$, $GH^* \leftarrow K_1$ (a single node);

Begin

1. If $N_{GH^*} < O_{GH^*}$ then $GH^* \leftarrow GH^* \times m_h$, $N_{GH^*} \leftarrow N_{GH^*} \times m_h$, $O_{GH^*} \leftarrow O_{GH^*}/m_h$ and $h \leftarrow h+1$ else go to Step 3.
2. If $N_{GH^*} < O_{GH^*}$ then $GH^* \leftarrow GH^* \times m_t$, $N_{GH^*} \leftarrow N_{GH^*} \times m_t$, $O_{GH^*} \leftarrow O_{GH^*}/m_t$, $t \leftarrow t-1$, and go to Step 1.
3. If $O_{GH^*} \leq 2\sum_{i=1..r}(m_i-1) - m_j$, where $m_j = \max\{m_1, m_2, \dots, m_r\}$ then $f \leftarrow O_{GH^*} - 1$ and go to Step 8.
4. $O_{GH^*} \leftarrow 1$, $N_{GH^*} \leftarrow \prod_{i=1..r} m_i$, $h \leftarrow 1$, and $t \leftarrow r$, $GH^* \leftarrow K_1$.
5. If $O_{GH^*} \times m_h \leq 2\sum_{i=1..r}(m_i-1) - m_j$ then $GH^* \leftarrow GH^* \times m_h$, $O_{GH^*} \leftarrow O_{GH^*} \times m_h$, $N_{GH^*} \leftarrow N_{GH^*}/m_h$ and $h \leftarrow h+1$.
6. If $O_{GH^*} \times m_t \leq 2\sum_{i=1..r}(m_i-1) - m_j$ then $GH^* \leftarrow GH^* \times m_t$, $O_{GH^*} \leftarrow O_{GH^*} \times m_t$, $N_{GH^*} \leftarrow N_{GH^*}/m_t$, $t \leftarrow t-1$, and go to Step 5.
7. $f \leftarrow 2\sum_{i=1..r}(m_i-1) - m_j - 1$.
8. Output the GH^* , f , O_{GH^*} and N_{GH^*} .

End DECOMP_GH

After applying DECOMP_GH on a $GH(m_1, m_2, \dots, m_r)$, a GH^* , f , O_{GH^*} and N_{GH^*} can be determined. Obviously, all GH^* s ($HC(GH^*)$) in a $GH(m_1, m_2, \dots, m_r)$ are disjoint. In the following, the set of all $HC(GH^*)$ s in a $GH(m_1, m_2, \dots, m_r)$ is denoted by $S(HC^*)$.

Lemma 5. There exists a fault-free σ -zero cycle in

the $S(HC^*)$ of a $GH(m_1, m_2, \dots, m_r)$. In other words, every node in a σ -zero cycle is fault-free.

Proof. Applying DECOMP_GH on a $GH(m_1, m_2, \dots, m_r)$, $O_{GH^*} \leq N_{GH^*}$ and $f = \min\{O_{GH^*} - 1, 2\sum_{i=1..r}(m_i-1) - m_j - 1\}$. Thus, at least one $HC(GH^*)$ in the $S(HC^*)$ is fault-free and every node in it is fault-free by the fault-free cycle definition. $\square$

4 Diagnosis Algorithm

This section first presents the proposed diagnosis algorithm (GH_PESDIAG) and then reveals the proof of its correctness and the time complexity analysis. Before describing the GH_PESDIAG, some terms should be defined first. Let $HC(GH^*)$ be a σ -nonzero cycle in the $S(HC^*)$ of a $GH(m_1, m_2, \dots, m_r)$. Then, $HC(GH^*)$ is σ -guarded (σ -unguarded) if $HC(GH^*)$ is adjacent (not adjacent) to any σ -zero cycle. Moreover, each node in a σ -guarded (σ -unguarded) cycle is σ -guarded (σ -unguarded).

4.1. Algorithm GH_PESDIAG

The statements of the algorithm GH_PESDIAG is presented.

Algorithm GH_PESDIAG

Input: A syndrome σ on a $GH(m_1, m_2, \dots, m_r)$ produced by at most $f = \min\{O_{GH^*} - 1, 2\sum_{i=1..r}(m_i-1) - m_j - 1\}$ faulty nodes, $m_j = \max\{m_1, m_2, \dots, m_r\}$.

Output: A sequence (V_0, V_1, V_2) , where V_0 is the set of nodes that are identified to be fault-free; V_1 is the set of nodes that are identified to be faulty, and $V_1 \cup V_2$ is the isolated set that contains all faulty nodes. Initially, all nodes of the $GH(m_1, m_2, \dots, m_r)$ are set as ‘undiagnosed’.

Initialization: $V_0 \leftarrow \{\}$, $V_1 \leftarrow \{\}$, $V_2 \leftarrow \{\}$;

Begin

1. Apply DECOMP_GH.
2. Each node in a $HC(GH^*)$ diagnoses its two neighbors.
3. Identify all σ -zero nodes to be fault-free. Add all σ -zero nodes into V_0 .
4. While there is a ‘fault-free’ node u that is adjacent to a σ -guarded node v , diagnose v as ‘fault-free’ or ‘faulty’ according as $\sigma(u, v) = 0$ or 1. Add all fault-free nodes into V_0 and all faulty nodes into V_1 .
5. While there is a ‘fault-free’ node u that is adjacent to a σ -unguarded node v , diagnose v as ‘fault-free’ or ‘faulty’ according as $\sigma(u, v) = 0$ or 1. Add all fault-free nodes into V_0 and all

faulty nodes into V_1 .

6. While there is a ‘fault-free’ node u that is adjacent to a ‘undiagnosed’ node v , diagnose v as ‘fault-free’ or ‘faulty’ according as $\sigma(u, v) = 0$ or 1. Add all fault-free nodes into V_0 and all faulty nodes into V_1 .
7. If $|V_1| = f$, then diagnose all ‘undiagnosed’ nodes as ‘fault-free’ else diagnose all ‘undiagnosed’ nodes as ‘doubtful’. Add all fault-free nodes into V_0 and all doubtful nodes into V_2 .

End GH_PESDIAG

4.2. Correctness

The correctness of the GH_PESDIAG is addressed by the following theorem.

Theorem 6. After running the GH_PESDIAG on a $\text{GH}(m_1, m_2, \dots, m_r)$ with at most $f = \min\{O_{\text{GH}^*} - 1, 2\sum_{i=1..r}(m_i - 1) - m_j - 1\}$ faulty nodes, a node is fault-free if it is diagnosed as ‘fault-free’; a node is faulty if it is diagnosed as ‘faulty’, and at most one node is diagnosed as ‘doubtful’.

Proof. According Lemma 5, there exists a fault-free σ -zero cycle in the $S(\text{HC}^*)$ of a $\text{GH}(m_1, m_2, \dots, m_r)$. Thus, every node in a σ -zero cycle is fault-free and the algorithm can correctly start and terminate. Additionally, the assumptions of the PMC model assure that, upon completion of the algorithm, $V_0 \subseteq V(\text{GH}(m_1, m_2, \dots, m_r)) - F$ and $V_1 \subseteq F$. Therefore, all the faulty nodes have been isolated to within the set $V_1 \cup V_2$. It remains yet to prove that $|V_2| \leq 1$ and $|V_1| + |V_2| \leq f$. Clearly, the two inequalities hold for the case when $|V_1| = f$. So, assume $|V_1| \leq f - 1$. Let us have a look at what happens just after Step 6 of the GH_PESDIAG has been executed. One of the following three possibilities may take place.

Case 1. All nodes have been diagnosed. Then $|V_2| = 0$ and hence $|V_1| + |V_2| \leq f < 2\sum_{i=1..r}(m_i - 1) - m_j$, $m_j = \max\{m_1, m_2, \dots, m_r\}$.

Case 2. There is exactly one undiagnosed node, say u . Then $V_2 = \{u\}$. So, $|V_2| = 1$ and $|V_1| + |V_2| \leq f < 2\sum_{i=1..r}(m_i - 1) - m_j$, $m_j = \max\{m_1, m_2, \dots, m_r\}$.

Case 3. There were exactly two undiagnosed node say u and v . Note that this would imply $f \geq |N(\{u, v\})|$ and $|N(\{u, v\})| \geq 2\sum_{i=1..r}(m_i - 1) - m_j$. Thereby we could conclude $f \geq 2\sum_{i=1..r}(m_i - 1) - m_j$. It is a contradiction. $\square$

4.3. Time complexity

This section analyzes the time complexity of the GH_PESDIAG.

Theorem 7. The time complexity of the GH_PESDIAG is $O(N \log N)$, where $N = m_1 \times m_2 \times \dots \times m_r$ is the total number of nodes in the $\text{GH}(m_1, m_2, \dots, m_r)$, where $m_i \geq 2$ and $1 \leq i \leq r$.

Proof. Step 1 takes $O(r)$ time; Step 2 takes $O(1)$ time; Step 3 takes $O(N^{1/2})$; Steps 4–6 takes $O(N \log N)$ time, and Step 7 takes $O(1)$. Therefore, the time complexity of the GH_PESDIAG is $O(N \log N)$. $\square$

5 Conclusions

This work first shows that a $\text{GH}(m_1, m_2, \dots, m_r)$ is Hamiltonian and derive a algorithm for decomposing the $\text{GH}(m_1, m_2, \dots, m_r)$ into proper subcubes which have the same structure. Finally, a fast pessimistic diagnosis algorithm for diagnosing a $\text{GH}(m_1, m_2, \dots, m_r)$ is provided by exploiting its cycle decomposition properties. Even though that a GH is a super set of a hypercube, comparing with the CD-DIAG proposed by Yang, the new provided algorithm is not slower than Yang’s algorithm and can run in $O(N \log N)$ time.

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